



**A NOTE ON SHALLOW WATER OCTOCORALLIA FROM GULF OF MANNAR,
SOUTH EAST COAST OF INDIA**

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ABSTRACT

Gulf of Mannar is one of the world's richest marine biospheres and occupies an area of 10,500sq km, 21 coral reef islands and small patch reefs are present between Lat. 80° 47'N; Long. 78° 12'E to Lat. 90° 15' N; Long. 79° 14' E. About 3600 species of fauna and flora have been reported from this area; out of which 351 species of octocorals reported from India among them 89 species of Octocorals have been reported from Gulf of Mannar. The present study revealed the diversity of shallow water octocorallia (Helioporacea, Alcyonacea, Pennatulacea, Gorgonacea) in the Gulf of Mannar Marine National Park. The sampling were made at 74 study sites, so far 13 sites in Thoothukudi range, 4 sites in Vembar range, 28 sites in Keelakari range and 29 sites in Mandapam range with the help of SCUBA diving up to 25 m depth. The present study was reported 62 species belonging to 23 genera, 12 families and 3 orders in the Gulf of Mannar. Of which the Alcyonacea was a dominant group followed by Gorgonacea and Pennatulacea at all the study sites.

Key Words: Alcyonacea, Gorgonacea, Pennatulacea, Soft coral, Octocorallia, Gulf of Mannar.



RECOVERY STATUS OF *PORITES COMPRESSA* CORAL IN TURBID
ENVIRONMENT FROM MARINE NATIONAL PARK, GULF OF KACHCHH

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ABSTRACT

The Gulf of Kuchchh is one of the world richest marine biospheres, occupies an area of 7350 sq. km and 42 coral reef islands are present, these islands possess a very interesting heterogeneous group of fauna and flora. The study site is characterised by a strong tidal variation and the reef communities are capable of higher exposure time during negative tides. The present experimental study was conducted between February 2015 to July 2015 (six month). We collected three *Porites compressa* samples from three different donor colonies at Pirotan Island. The donor colonies were revisited on each month to observe the recovery status. The sedimentation rate ranged between 14.7 to 94.3 mg/cm²/day. The observed recovery of donor colony was 35mm in first colony, 28mm in second colony and 17mm in the third colony at Pirotan Island. The surface water temperature and sedimentation are vital for the normal growth of corals in the natural and artificial environment.

Key Words: coral reef, *Porites compressa*, Marine National Park, Gulf of Kachchh.



A NOTE ON SHALLOW WATER OCTOCORALLIA
FROM NICOBAR ISLANDS, INDIA

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ABSTRACT

Nicobar Islands are situated in the southeast region of Bay of Bengal, between 6⁰-10 N Lat. and 92⁰ - 94⁰ E Long. There are 22 islands in Nicobar group, of which 12 are inhabited. Present study revealed the diversity of shallow water soft coral octocorallia (Helioporacea, Alcyonacea, Pennatulacea, Gorgonacea) in Nicobar group of Islands. The sampling made at 11 study sites (Malacca - MA, Kakanna - KA, Tamaloo - TA, Parka - PA, Lapathi - LA, Nancowry - NA, Champian - CH, Kamorta - KM, Munak - MU, Trinket - TR, Great Nicobar - GN) with help of SCUBA diving to the depth of 5 to 30 m. A total of 27 species of octocorals under 23 genera, 13 families were recorded from the Nicobar group of Islands. A maximum number of species were recorded from TR (14 sp.) and minimum at TA (5 sp.). The Menhinick Diversity index was recorded as highest at PA (1.5) while LA and CH (0.8) represented the lowest diversity of Octocorals. The Shannon - Weiner Diversity index was maximum in KM (1.24) and minimum in MU (0.57). The Simpson's Diversity indices, KM (0.68) was noted maximum and MU (0.29) minimum and the Pielou's Evenness index for species community ranged from 0.59 at MU to 0.91 at LA. A maximum number of individuals and percentage of cover reported in TR and minimum in TA. Of which the Alcyonacea was dominant group followed by Gorgonacea, Helioporacea and Pennatulacea at all the study sites which is contributed with result obtained based on Principal Component analysis, De-Trended analysis and Ternary plot. Bray-Curtis cluster analysis resulted 75% similarity between the study sites in Nicobar Islands.

Key Words: Soft coral, Octocorallia, diversity, Andaman and Nicobar Islands.